

```
//
PImage neuesBild =
createImage(originalbild.width,originalbild.height,RGB);
```

```
//
double rot   = 0.0;
double gruen = 0.0;
double blau  = 0.0;
```

```
//
int xx = x - halb;
int yy = y - halb;
```

```
//
int laenge = filter.length;
int halb   = laenge / 2;
```

```
//
for (int i = 0; i < laenge; i++) {
```

```
//
for (int j = 0; j < laenge; j++) {
```

```
//
for (int x = halb; x < originalbild.width - halb; x++){
```

```
//
for (int y = halb; y < originalbild.height - halb; y++) {
```

```
//
rot   += filter[i][j] * red(pixelAlt[xx+i][yy+j]);
gruen += filter[i][j] * green(pixelAlt[xx+i][yy+j]);
blau  += filter[i][j] * blue(pixelAlt[xx+i][yy+j]);
```

```
//
if(rot < 0.0)      rot   = 0.0;
if(rot > 255.0)    rot   = 255.0;
if(gruen < 0.0)    gruen = 0.0;
if(gruen > 255.0)  gruen = 255.0;
if(blau < 0.0)     blau  = 0.0;
if(blau > 255.0)   blau  = 255.0;
```

```
//
pixelNeu[x][y] = color((int) rot, (int) gruen, (int) blau);
neuesBild.set(x,y,pixelNeu[x][y]);
```

```
return neuesBild;
```

```
public PImage faltung(PImage originalbild, double[][] filter) {
```

```
}
```

```
}
```



```
}

}

}
```